CLASS TEST

CO-2 AT-3

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① Image Enhancement in the Spatial Domain

The image enhancement in the spatial domain improves the visuals quality of an image by directly modifying the image pixel values. It is mainly used to improve brightness, contrast, remove noise, and highlight important details.

a) Gray-Level Transformation

Gray level transformation changes the intensity values of pixels.

Types:
Image Negative, Log transformation, powerlaw

b) Histogram processing

A histogram processing represent the distribution of pixel intensities

Techniques: Histogram equalization, make histogram matching

c) Spatial Filtering

Spatial filtering modifies a pixels using mask (Kernel)

Types:

* Smoothing Filters (mean, median)
* Sharpening Filters (Laplacian, Sobel)

② Image Smoothing vs Image Sharpening

Image Smoothing	Image Sharpening
* Remove noise	* Enhance edges
* Blur image slightly	* Increase details
* uses Low-pass filters	* use high-pass filters
* Reduces high frequency Component	* Enhance high frequency components

Spatial Smoothing Filters

They replace each pixel with the average or median of neighboring pixels.

Types: Mean filter, median Filter, Gaussian Filter

Advantages:

* Removes Gaussian and salt-and-pepper noise
* Produce smoother image

Sharpening Filtering

Highlight sudden intensity changes (edges)

Types:

* Laplacian Filter
* Sobel Filter
* Prewitt Filter

5) Image Enhancement in the Frequency Domain

A frequency domain Enhancement modifies the image after converting it into the frequency domain using the Fourier Transform (FFT)

Steps:

i) Convert image using FFT

ii) Apply frequency filter

iii) perform inverse FFT

Low pass Filter

working:

allow low frequencies and blocks high frequencies

Effects:

* Remove noise
* produces blurred image

High Pass Filter

working:

allow high frequencies and suppresses low frequencies.

Effects:

* Enhance edges
* Increases sharpness

Comparison

Low-Pass Filter	High-Pass Filter
Remove noise	Enhances edges
Smooth image	sharpens images
produces blur	produces details image

Image Segmentation

Image segmentation divides an image into meaningful region for easier analysis

Technique	Working Principle
* Point Detection	* Detects isolated pixels
* Line Detection	* Detects straight lines
* Edge Detection	* Detects intensity changes
* Thresholding	* Separates object using intensity values

Best Technique:

→ Edge Detection - Accurate object boundaries

→ Thresholding - Simple images with clear intensity differences

→ Region-Based segmentation - complex images

5) Edges Detection, Edge linking and Boundary Detection

Edge Detection

Detect Sudden changes in intensity that represent object boundaries.

Operators
* Sobel
* prewitt
* canny
* Roberts

Importance:
* Object recognition
* shape analysis
* medical diagnosis

Edge Linking

connects broken edges segments into continous boundaries

Identifies the outer boundary of an object of an object after segmentation.

Importance
* Measure object size
* Shape recognition
* Feature extraction